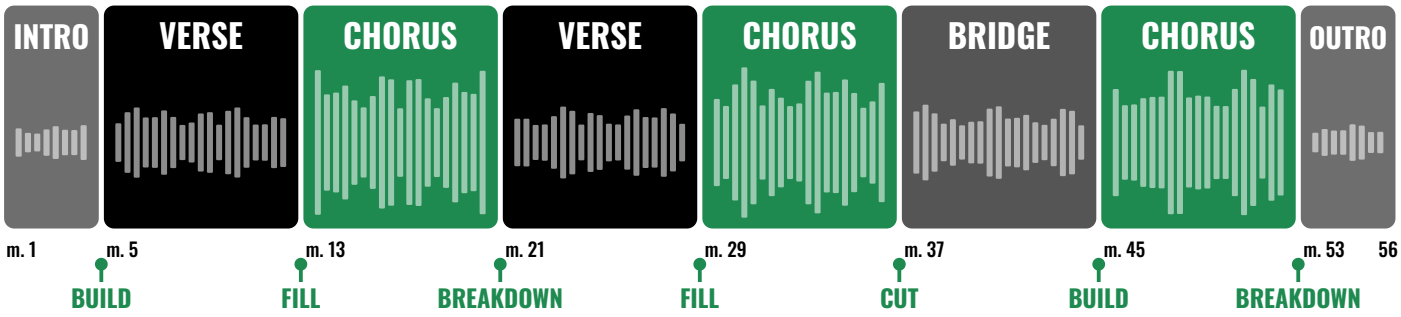
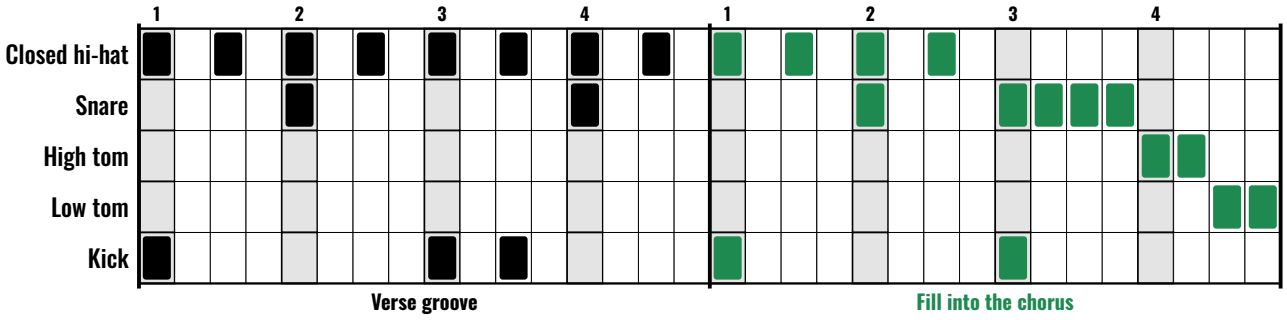
HUDSON BEAT





DRUMS OVER STEMS IN LOGIC PRO

STEMS & GROOVES

DMP1 · Unit 2 · listen first · fills, breakdowns, cuts, builds · gain staging

GRADES 9–12 · 6 DAYS · 28 PAGES + 22-SLIDE DECK

STEMS & GROOVES

How to use this resource

START HERE

Teacher

WHAT THIS IS ·

A complete 6-day drum unit for high school music technology. Students write drums for music that already exists: a stem, a recording with the drums taken out. In Part 1, Foundation, they write two contrasting patterns, A and B, and place them where the sections change in a 16-measure stem. In Part 2, Full Song, they map every section change in a longer stem and announce each one with a fill, a breakdown, a cut or a build. The mixing concept is gain staging: nothing in the red.

This is Unit 2 of my Digital Music Production 1 course, after Unit 1, Beat From Nothing (a separate resource), where students learned Drum Machine Designer and the Step Sequencer.

QUICK START ·

1. Read the overview and pacing (pages 3 and 8) and the teacher notes (pages 12–14).
2. **Get your stems** (pages 5–6): a few 16-measure Foundation stems and a few Full Song stems. None are included.
3. Program the four example transitions (page 26) and cue up In the Air Tonight (Phil Collins) on YouTube or Spotify.
4. Print the student pages (pages 15–24), one set per student.

THE INTRO DECK ·

Every slide has a day: Day 1 (slides 1–9), Day 2 (slides 10–12), Day 3 (slides 13–15), Day 4 (slides 16–17), Day 5 (slides 18–20) and Day 6 (slides 21–22). The day plans name the slide for each block. Every slide has speaker notes.

Fonts: the deck uses Oswald and Source Sans 3 (free on Google Fonts, built into Google Slides). If PowerPoint or Keynote shows a different font, install the two fonts or present the PDF copy, which shows the intended look.

WHAT'S INSIDE ·

PART	PAGES	USE
Teacher pages	pages 2–14	How to use, overview, setup, the stems (find or make them), Logic reference, pacing, day plans, teacher notes
Student pages	pages 15–24	Day 1 post, the four transitions, gain staging, section map, the project sheet (4 pages), rubric, name that transition
Answer keys + examples	pages 25–27	Worked A and B patterns, the four transitions and quiz answers, a mapped full song
Intro deck (separate file)	deck	22 slides, .pptx + PDF, speaker notes on every slide

NO STEMS INCLUDED

No stems, audio or Logic sessions are included in this kit. You supply the stems (recordings with the drums taken out): make your own in Logic with the recipe on page 6, or use audio you have the right to use in class (page 5).

ABOUT THE SONGS

Songs are named by title and artist only. Play them from your own YouTube or Spotify access; no audio, video or lyrics are included.

MAKE IT FIT YOUR CLASS ·

- **File names:** my requirements say "Your Name - Foundation" and "Your Name - Full Song"; my checklist says "Your Name - Foundations" for both. Tell students which names you want.
- **Grading:** my rubric, five categories scored 2–5, /25 (page 23).
- **More accessible / more advanced:** variations on page 14.
- **Feedback board:** I use Padlet for the three feedback questions; any class board works.

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STEMS & GROOVES

Unit overview

OVERVIEW

DMP1 · Unit 2

WHAT STUDENTS MAKE ·

Two drum tracks written to fit stems the teacher supplies. **Foundation:** an A pattern and a contrasting B pattern, at least 3 layers each, placed where the sections change in a 16-measure stem. **Full Song:** drums for a full-length stem with a deliberate transition at every section change, at least two types, mixed so the drums serve the stem.

WHAT STUDENTS LEARN ·

AABA form through doing — not naming. How drums change when a section changes. Why musical continuity matters. How to listen to a piece of music and respond to it. That mixing decisions (volume, panning) are musical decisions, not technical ones.

WHY THIS UNIT LANDS HERE ·

The stems project is the most musically sophisticated thing in the early curriculum. It removes the blank-page problem while requiring genuine musical thinking. Students hear what a complete track sounds like and have to fit into it — that is a listening and responding skill that transfers to every project for the rest of the year.

MIXING CONCEPT: GAIN STAGING ·

The stem tracks have varying levels, and students notice at once when their drums are too loud or too quiet. The rule: “Never push anything into the red. It is almost always better to turn other things down.”

GOALS ·

- Listen to a piece of music and identify where its sections change
- Write two contrasting drum patterns that serve the music they accompany
- Use transitions (fills, breakdowns, cuts, builds) to signal section changes
- Apply gain staging so the drum track sits correctly in the overall mix
- Experience what it means to respond musically to something that already exists

AT A GLANCE ·

Grades	9–12 (no music reading needed)
Length	6 periods of about 53 minutes: learning day, three work days, submit, present
Software	Logic Pro: Drum Machine Designer, pattern regions, Step Sequencer (16 and 64 steps), arrangement markers, volume and pan, export
You supply	The stems (none included): find or make them, pages 5–6
Product	Two projects, Foundation and Full Song, each an MP3 + compressed project; three feedback questions
Assessment	My 5-category rubric, scored 2–5, /25 (page 23)

STANDARDS ·

Supports the National Core Arts Standards for Music (Music Technology, high school): developing musical ideas (MU:Cr2.1), refining work (MU:Cr3.1) and presenting work (MU:Pr6.1).

FROM MY TUTORIAL

“Drums in songs tend to remain relatively consistent. That is what ties everything together and makes it sound like one song.”

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STEMS & GROOVES

Setup and requirements

SETUP

Teacher

YOU NEED ·

- **A Mac with Logic Pro** for each student (Logic Pro X 10.5 or later, for the Step Sequencer and pattern regions), and **headphones**.
- **Stems**: a Foundation folder (several 16-measure stems, sections of 4 measures) and a Full Song folder (several full-length stems with clear sections), on your class page. None are included: pages 5–6.
- **Projector and speakers** for the deck, the stems and In the Air Tonight (Phil Collins), from your own YouTube or Spotify access.
- **The four example transitions**, programmed once in a Logic project from page 26 (10 minutes), for Days 1 and 3.
- **Printed student pages**, one set each: the Day 1 post, the four transitions, gain staging, the section map, the project sheet and rubric, name that transition.
- **A place to submit** an MP3 and a .zip, and a class feedback board for the three questions (I use Padlet).

A MIDI KEYBOARD? ·

Not needed. Every step in this unit is clicked into the Step Sequencer.

SKILLS FROM UNIT 1 THIS UNIT USES

Drum Machine Designer, pattern regions, the Step Sequencer (16 and 64 steps), volume and pan in the Mixer, and export. Students new to them need about 10 extra minutes on Day 1 with the Logic reference (page 7).

BEFORE DAY 1 ·

- [ ] Put the Foundation stems on your class page, each named with its tempo (for example “Stem 1 - 96 BPM”).
- [ ] Open one stem in Logic yourself: set the tempo, drop it at bar 1, check the click lines up.
- [ ] Program the four example transitions (page 26).
- [ ] Test the class speakers; cue up the song.
- [ ] Print the student pages; set up the submission spot and the feedback board.

BEFORE DAY 3 ·

- [ ] Put the Full Song stems on your class page.
- [ ] Optional scaffolding: pre-label the sections with arrangement markers for students who need it.

HOW STUDENTS OPEN A STEM

New empty project. Set the project tempo to the stem’s BPM. Drag the audio file into the Tracks area so it starts at bar 1. Turn on the metronome (K): the click should line up with the music. Then add the drum track.

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STEMS & GROOVES

The stems: what you supply

STEMS

Teacher

WHAT IS (AND IS NOT) IN THIS KIT

No stems, audio or Logic sessions are included in this kit. You supply the stems (recordings with the drums taken out): make your own in Logic with the recipe on page 6, or use audio you have the right to use in class (page 5). My own classroom stems come from other artists, so I cannot include them.

WHERE TO GET STEMS ·

1. Make your own (safest). Build a short backing track in Logic with no drums and bounce it to an audio file. The recipe on page 6 makes one Foundation stem and one Full Song stem in about 30 minutes.

2. A multitrack library (check its terms first). I source new tracks from the multitrack library at cambridge-mt.com/ms/mtk. Before you use it, read the library's own terms and each track's: some libraries allow personal mixing practice but not handing files to students. If the terms allow it, leave out (or mute) the drum tracks and bounce the rest into one stem.

3. Your own recordings: your school band, a colleague, or you, playing without a drummer.

4. Royalty-free music whose license allows you to edit it and share it with students, if you can get a version without drums.

Only use audio you have the right to use and share in class. Do not hand out stems pulled from commercial songs.

WHAT MAKES A GOOD STEM ·

- **No drums.** Everything else can stay: bass, chords, melody, vocals.
- **A steady tempo you know.** Put the BPM in the file name so students can set Logic to match.
- **Clear sections.** Energy that changes where the sections change: instruments in or out, louder or softer.
- **Starts on the downbeat of bar 1**, with no pickup or silence before it, so the grid lines up.
- **Foundation:** 16 measures, sections of 4 measures (A or B), as in my Day 1 post.
- **Full Song:** 2 to 4 minutes with an intro, verses, choruses and ideally a bridge.
- **WAV or MP3**, one file per stem.

TWO FOLDERS ON YOUR CLASS PAGE ·

FOLDER	USED	WHAT
Foundation	Days 1–2	Several 16-measure stems; students pick one that catches their ear
Full Song	Days 3–6	Several full-length stems; each student uses a new one

My project sheet calls the stem "a real professional recording". With stems you made, that sentence means your stem.

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STEMS & GROOVES

Make your own stems in Logic

STEMS

Recipe · 30 min

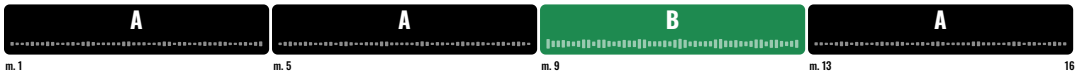
Two original stems, C major, white keys only. Every chord is a white-key chord of C major: I, iii, IV, V and vi. Change the chords, sounds or tempo to make more; keep the section lengths.

1. New project, tempo 96 BPM, 4/4. Software Instrument tracks: a piano, a bass, a pad and a simple synth. No drum track.
2. Turn on Cycle over 4 measures. With the Pencil in the piano roll, enter one chord per measure: three notes stacked, all white keys (the chord names below). The bass plays each chord's root, low, as a whole note.
3. Copy the 4-measure progression (Option-drag) to build each section; mute or delete instruments so each section has the instruments in the table. More instruments = more energy.
4. Add arrangement markers for your own answer key. Then File › Bounce › Project or Section, WAV or MP3, starting at bar 1.
5. Name the file with its tempo, for example "Foundation 1 - 96 BPM", and post it in the right folder.

FOUNDATION STEM · 16 MEASURES · A A B A ·

SECTION	MEASURES	CHORDS (ONE PER MEASURE)	INSTRUMENTS
A	1–4	I – vi – IV – V (C Am F G)	Piano chords, Bass
A	5–8	I – vi – IV – V (C Am F G)	Piano chords, Bass
B	9–12	IV – V – iii – vi (F G Em Am)	Piano chords, Bass, Pad, Synth plays the top note of each chord
A	13–16	I – vi – IV – V (C Am F G)	Piano chords, Bass

16 measures at 96 BPM = 0:40 (Foundation).



FULL SONG STEM · 56 MEASURES ·

SECTION	MEASURES	CHORDS (ONE PER MEASURE)	INSTRUMENTS
Intro	1–4	I – vi – IV – V (C Am F G)	Pad
Verse	5–12	I – vi – IV – V (C Am F G)	Piano chords, Bass
Chorus	13–20	IV – I – V – vi (F C G Am)	Piano chords, Bass, Pad, Synth plays the top note of each chord
Verse	21–28	I – vi – IV – V (C Am F G)	Piano chords, Bass
Chorus	29–36	IV – I – V – vi (F C G Am)	Piano chords, Bass, Pad, Synth plays the top note of each chord
Bridge	37–44	vi – IV – vi – V (Am F Am G)	Pad, Bass
Chorus	45–52	IV – I – V – vi (F C G Am)	Piano chords, Bass, Pad, Synth plays the top note of each chord
Outro	53–56	IV – V – I – I (F G C C)	Pad

56 measures at 96 BPM = 2:20 (Full Song).

Chord spellings: I = C E G · ii = D F A · iii = E G B · IV = F A C · V = G B D · vi = A C E. Worked drum answers for both stems: pages 25–27.

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STEMS & GROOVES

Logic quick reference

SETUP

Teacher

Everything students need in Logic for this unit, in one table. Demonstrate once (slides 9 and 15), then let students work.

Open a stem	New empty project. Set the tempo (double-click it in the control bar) to the stem's BPM. Drag the audio file into the Tracks area at bar 1.
New drum track	Track › New Software Instrument Track; in the Library (Y) choose any kit under Electronic Drum Kit. This loads Drum Machine Designer.
Arrangement markers	Turn on Global Tracks (G). In the Arrangement track, click + to add a marker, drag its edges to fit the section, and rename it (A, B, Verse…).
Pattern region	Control-click (right-click) in the drum track at the start of a section and choose Create Pattern Region. Double-click it to open the Step Sequencer.
Place hits	Click steps to turn hits on. Each row is one sound; add rows for more sounds.
64-step mode	In the Step Sequencer, set the pattern length to 64 steps: four measures in one pattern, so a transition can sit in its last measure.
Copy across	Option-drag a region to copy it, or select it and press Command-R to repeat it.
Cycle	Press C and drag the cycle area over a section to loop it while you work.
Volume and pan	Open the Mixer (X). Click the disclosure triangle on the Drum Machine Designer track to show each drum's own fader and pan knob.
Clipping	The peak display above each fader turns red when that track clips. Turn something down, then click the display to reset it.
Export	File › Bounce › Project or Section (Command-B), MP3. In Finder, Control-click the .logicx project and choose Compress.

THE ORDER THAT WORKS

Listen twice, markers, then drums. Groove first (kick, snare, closed hi-hat), then layers, then transitions, then the mix.

VERSION NOTE

Menu names are from current Logic Pro versions; older versions may differ slightly. Logic Pro is a trademark of Apple Inc., named only to describe what the unit needs. This resource is not affiliated with Apple.

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STEMS & GROOVES

Pacing guide · 6 days

PACING

Teacher

Six class periods of about 53 minutes, as in my course plan: a learning day, three work days, a submission day and a presenting day. Foundation first, then Full Song. Each day's full plan is on pages 9–11.

DAY	FOCUS AND GOAL	MINUTE BLOCKS	DECK
DAY 1 Learning	**Learning + Foundation start** Every student has a stem downloaded, sections labeled, and an A pattern started.	0–5 · The stem, no drums 5–15 · Introduce the project 15–25 · Listen twice 25–48 · Markers + A pattern 48–53 · Save	Intro deck slides 1–9
DAY 2 Work day	**Working: Foundation** Foundation complete. Both patterns placed correctly. Mix adjusted.	0–5 · Quick check 5–45 · A, B, place 45–53 · Gain staging	Intro deck slides 10–12
DAY 3 Work day	**Transition workshop + Full Song start** At least two transitions built. Students understand the difference between the four types.	0–15 · Transition workshop 15–48 · Map + build 48–53 · Save	Intro deck slides 13–15
DAY 4 Work day	**Working: Full Song** All section changes have deliberate transitions. Mix adjusted. Project ready to submit.	0–53 · Full working day	Intro deck slides 16–17
DAY 5 Submit	**Submission + presentation prep** All projects submitted before Day 6.	0–20 · Finish + submit 20–53 · Early presentations	Intro deck slides 18–20
DAY 6 Present	**Presenting** Every student plays their full drum track alongside the stem; every student hears the professional skill they just used.	0–5 · Last submissions 5–53 · Presentations	Intro deck slides 21–22

STUDENT PAGES BY DAY ·

Day 1	Day 1 post (p. 15), project sheet (pp. 19–22), section map Part 1 (p. 18)
Day 2	Gain staging (p. 17)
Day 3	The four transitions (p. 16), section map Part 2 (p. 18), name that transition (p. 24)
Days 4–6	Checklist (p. 22) and rubric (p. 23)

MY LINE FOR THE WHOLE UNIT

"Never push anything into the red. It is almost always better to turn other things down." And before anything: listen first, then respond.

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STEMS & GROOVES

Day-by-day plans · Days 1 and 2

DAYS 1–2

Foundation

DAY 1 · LEARNING + FOUNDATION START

Learning · Intro deck slides 1–9

Goal: Every student has a stem downloaded, sections labeled, and an A pattern started.

0–5 **THE STEM, NO DRUMS** [slides 1–2] · Slides 1–2. Play 60 seconds of one of the stem tracks without the drums. Ask: what is this music doing? Where does the energy change? Do not give answers yet.

5–15 **INTRODUCE THE PROJECT** [slides 3–6] · Slides 3–6. Explain Foundation and Full Song. Show the four transition types with audio examples (In the Air Tonight, then the four you programmed from page 26). Keep this brief — students will understand transitions better once they have built patterns.

15–25 **LISTEN TWICE** [slide 7] · Slide 7. Students download their stem track and listen twice before touching Logic. Circulate and ask every student: where are your sections? They should be able to tell you (section map, page 18).

25–48 **MARKERS + A PATTERN** [slides 8–9] · Slides 8–9: the worked example, then a short demo. Students add arrangement markers and begin building their A pattern. Prompt: does your groove fit the feel of the section it lives in?

48–53 **SAVE** · Students save, named “Your Name - Foundation”. Quick check of the day’s goal as they pack up.

DAY 2 · WORKING: FOUNDATION

Work day · Intro deck slides 10–12

Goal: Foundation complete. Both patterns placed correctly. Mix adjusted.

0–5 **QUICK CHECK** [slide 10] · Slide 10. Ask two students to play their A pattern alongside the stem. Does it fit? Whole class listens and responds briefly.

5–45 **A, B, PLACE** [slide 11] · Slide 11. Students complete A pattern, build B pattern, and place both across the full stem. Teacher circulates. Key prompt for B section: play A, then play B. Can you feel the difference without looking at the screen?

45–53 **GAIN STAGING** [slide 12] · Slide 12. Students adjust volumes. Nothing should be clipping. Introduce the gain staging rule explicitly: the stem is the lead, the drums support it (student page 17).

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DAY 3 · TRANSITION WORKSHOP + FULL SONG START

Work day · Intro deck slides 13–15

Goal: At least two transitions built. Students understand the difference between the four types.

0–15 **TRANSITION WORKSHOP** [slide 13] · Slide 13. Show all four types with a real example: drum fill, breakdown, cut, build. Play each one twice and ask students to describe what it does to the energy. This is the most important instructional moment of the unit.

15–48 **MAP + BUILD** [slides 14–15] · Slides 14–15: a mapped full song and a 64-step demo. Students map every section change in their stem and begin adding transitions. Teacher circulates. Key question at every transition: why did you choose this type for this moment?

48–53 **SAVE** · Students save the new project, named “Your Name - Full Song”. Exit question: which transition type have you not tried yet?

DAY 4 · WORKING: FULL SONG

Work day · Intro deck slides 16–17

Goal: All section changes have deliberate transitions. Mix adjusted. Project ready to submit.

0–53 **FULL WORKING DAY** [slides 16–17] · Slides 16–17. Students complete all transitions and finalize mix. Common prompt: play the full track from beginning to end without stopping. Every section change should feel like it was meant to happen. If anything feels accidental, it needs work. Early finishers may try the optional reverb bus.

CIRCULATING PROMPTS (DAYS 1–4)

- ‘Where are your sections? Show me before you build.’
- ‘Play A, then play B. Can you feel the difference without looking?’
- ‘Why did you choose this type for this moment?’

WHAT TO LISTEN FOR AT EVERY TRANSITION

- A transition that works feels inevitable in hindsight. If it surprises you in a good way, it will surprise the listener too.
- The cut is the most powerful transition and the easiest to execute badly. Silence only works if the music earns it. Listen carefully before and after.
- Only the measure before the change moves; the groove around it stays. Drums in songs tend to remain relatively consistent. That is what ties everything together and makes it sound like one song.

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STEMS & GROOVES

Day-by-day plans · Days 5 and 6

DAYS 5–6

Submit + present

DAY 5 · SUBMISSION + PRESENTATION PREP

Submit · Intro deck slides 18–20

Goal: All projects submitted before Day 6.

0–20 **FINISH + SUBMIT** [slides 18–20] · Slides 18–20: both checklists and the rubric. Students who are not yet done finish and submit. Students who are done write their three Padlet feedback questions.

20–53 **EARLY PRESENTATIONS** · Early presentations if time allows. Otherwise this is a full working and finalization day.

DAY 6 · PRESENTING

Present · Intro deck slides 21–22

Goal: Every student plays their full drum track alongside the stem; every student hears the professional skill they just used.

0–5 **LAST SUBMISSIONS** · Any remaining submissions. Padlet questions posted.

5–53 **PRESENTATIONS** [slides 21–22] · Slides 21–22. Presentations. Each student plays their full drum track alongside the stem. Classmates respond to at least 2 projects. Close with the observation from page 12.

THE CLOSE (DAY 6)

"Your drums were written in response to music that already existed. That is what session musicians do. That is a professional skill."

From my course plan: name what students just did. **"You mixed a song."** They adjusted volumes and panned instruments without knowing that was mixing; naming it gives them ownership of a skill they already have.

THREE FEEDBACK QUESTIONS: EXAMPLES TO GET STUDENTS STARTED (DAY 5)

- "Does my B pattern feel different enough from my A pattern?"
- "Which of my transitions worked best, and did any feel accidental?"
- "Do my drums sit under the stem, or on top of it?"

Students write their own; these only model the kind of question that gets useful answers.

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WHAT THIS UNIT REINFORCES ·

This unit introduces the most important listening skill in the curriculum: responding to music that already exists. Students are not building in a vacuum — they have to hear what a professional recording is doing and fit into it. That constraint forces musical decisions that free-form building never does.

Foundation plants AABA thinking without naming it yet. Students are writing two contrasting patterns and placing them based on what they hear. This is form comprehension through action. Full Song introduces the concept of musical transitions — that the seam between sections is as compositionally significant as the sections themselves. This idea will echo through the EDM project in Unit 8 and every project after it.

The mixing concept introduced here is gain staging — the foundation of every mix decision students will make for the rest of the year. Frame it simply: nothing in the red, everything audible, the drums serve the stem. Students encounter this as a practical problem immediately because the stem tracks have varying levels and their drums often overwhelm them.

HOW THIS KIT DIFFERS FROM MY CLASSROOM VERSION

- My course plan describes an older version of this unit (Group 1: drums in AABA over a 16-bar stem; Group 2: full-song drums with volume, panning and reverb). This kit follows my newest project sheet, Foundation and Full Song. AABA lives in the 16-measure Foundation stem (sections of 4), and the reverb bus is an optional, ungraded extension (slide 17, page 17).
- My stems and my example Logic session are not included (third-party audio). Pages 5–6 replace them. The Day 1 post's "Check the example!" link becomes "Your teacher will play an example of each."
- My LMS is written as "your class page". Padlet is kept: it is the board I use; any feedback board works.
- Day plans: Days 1 and 3 end with a 5-minute save block (my notes stop at minute 48); Day 4 is one 53-minute block. Slide numbers are added to every block.
- Part 2 step 1: one closing parenthesis added. Otherwise the wording of my project sheet, rubric (/25), Day 1 post and teacher notes is kept as I wrote it, including the checklist's "Your Name - Foundations" for both projects (see page 2).
- My Day 1 post and project sheet say to label sections "A, B, A, B, etc.": that is an example of labelling, kept as written. The recipe stem and worked example in this kit are A, A, B, A (pages 6 and 25); your stems may put A and B in any order.
- Added to my day plans: the Day 3 exit question ("which transition type have you not tried yet?") and the Day 4 note that early finishers may try the optional reverb bus.
- New: the stems pages, the Logic reference, the four transitions page, gain staging, the section map, name that transition, the worked examples and answer keys, and the deck.

THE CLOSE (DAY 6)

"Your drums were written in response to music that already existed. That is what session musicians do. That is a professional skill."

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COMMON STUDENT STRUGGLES ·

Skipping the listening step: The most common failure is students who open Drum Machine Designer before they have listened to their stem. Enforce the pre-listening requirement. Walk around Day 1 and ask every student to tell you where their sections are before they create a single pattern region. If they cannot answer, they are not ready to build.

A and B patterns that are barely different: Students will often change one hi-hat hit and call it a B section. Push back during working days with a simple test: play A, then immediately play B. Can you feel the difference without looking? If not, it needs more contrast. A useful prompt: if A is busy and driving, what would the opposite feel like?

Transitions that are just louder versions of the pattern: Students who are not yet confident with drum fills will often just double the hi-hat and call it a transition. Show examples of all four transition types on Day 3 before they start Full Song. The cut and breakdown are the most underused and often the most effective.

Drums that drown out the stem: Students push faders up instinctively. Address this with a direct demonstration: play a student project with drums too loud, then pull the drum bus down until the stem is clearly the lead. Ask: which version sounds more like a real song? The answer is usually obvious. Introduce the rule here: if you can clearly hear every drum hit and the stem is getting lost, your mix is backwards.

Not knowing where sections are: Some stems have less obvious section changes. For students who are struggling, add arrangement markers to their Logic file before they download it so the sections are pre-labeled. This is appropriate scaffolding at this stage of the year.

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SUGGESTED VARIATIONS ·

More accessible: Pre-label all section changes in the stem file before distributing. Allow students to complete Foundation only and skip Full Song. Accept one transition type (fill only) rather than requiring two different types.

More advanced: Require all four transition types to be used at least once. Challenge students to write a third pattern (a C section) for any bridge or breakdown in the stem. Ask them to write a one-sentence rationale for every transition choice on their Padlet post.

MIXING CONCEPT: GAIN STAGING (DAY 2) ·

Gain staging lands here because students encounter it naturally — the stem tracks have varying levels and students immediately notice when their drums are too loud or too quiet relative to the music. Introduce the concept directly: 'Never push anything into the red. It is almost always better to turn other things down than to turn one thing up.' This principle will echo through every project for the rest of the year.

GRADING TIPS ·

- Five categories, each scored 2–5, /25 (page 23). Foundations and Technical Skill come from the Foundation project; Full Song from the second; Gain Staging and Documentation from both.
- Play A, then B, without looking: is the contrast obvious? Then play every section change of the full song: is there a deliberate transition at each, two types or more?
- Watch the meters on the loudest section: any red? Is the stem still the lead?
- Worked examples and answer keys: pages 25–27.

RESOURCES ·

- Stem tracks: the Foundation folder and the Full Song folder on your class page (update each year as needed); pages 5–6.
- cambridge-mt.com/ms/mtk: a multitrack library I use for sourcing new tracks (check each track's terms).
- The intro deck (.pptx and PDF), included, in place of my older video tutorials.

OBSERVATION NOTES ·

What worked, what to adjust, and anything worth remembering for next year.

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STEMS & GROOVES — DAY 1

Digital Music Production 1 · Unit 2

Today you are starting Part 1 of this project: Foundation. Follow this sequence in order.

Do not open Drum Machine Designer until you have completed Steps 1 and 2.

1. **DOWNLOAD** · Open your class page and download one stem track from the Foundation folder. Open it in Logic Pro.
   Pick something that catches your ear. You are going to spend a lot of time with this track.

2. **LISTEN — TWICE** · Play the full 16 measures from beginning to end. Do not touch anything. Then play it again.
   What is this music doing? Where does the energy change? Where does a new section feel like it starts?

3. **MAP YOUR SECTIONS** · Add arrangement markers in Logic to label every section change you hear: A, B, A, B, etc. Each section should be 4 measures long.
   You cannot write drums for music you have not listened to. This step is not optional.

4. **BUILD YOUR A PATTERN** · Create a pattern region at an A section. Build a groove that fits the feel of that section. Start with kick, snare, and hi-hat — then make it yours.
   Does this groove fit the music? Play it alongside the stem and listen.

5. **START YOUR B PATTERN** · Once your A pattern feels solid, create a new pattern region at a B section and start building something that contrasts with A.
   A and B should feel different. If you have to look at the screen to tell them apart, they are not different enough yet.

THE FOUR TRANSITION TYPES — YOU WILL USE THESE IN PART 2:

Drum fill — a rhythmic variation in the last 1–2 measures that signals a change is coming

Breakdown — strip to one element (kick only, hi-hat only) before the new section hits

Cut — complete silence for 1–2 beats before the next section drops

Build — add elements progressively in the last measure to increase energy into the change

Your teacher will play an example of each.

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STEMS & GROOVES

The four transitions

STUDENT

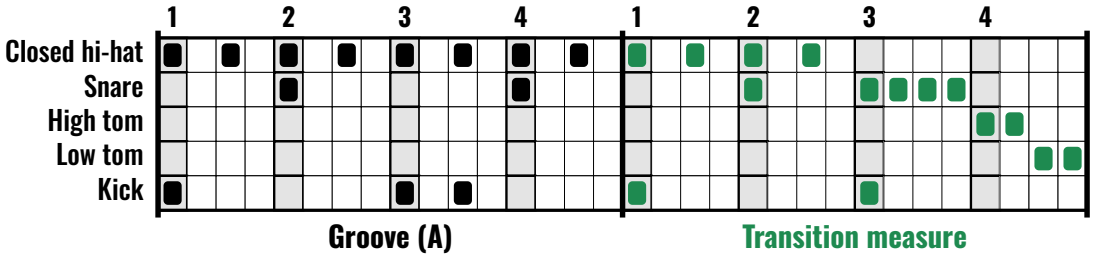
Reference

A transition is the moment between two sections where the drums signal that a change is coming. Each example shows one measure of the groove, then the transition measure (green) right before the new section. Each grid column is a sixteenth note; grey columns are the beats.

1 · DRUM FILL

a rhythmic variation in the last 1–2 measures that signals a change is coming.

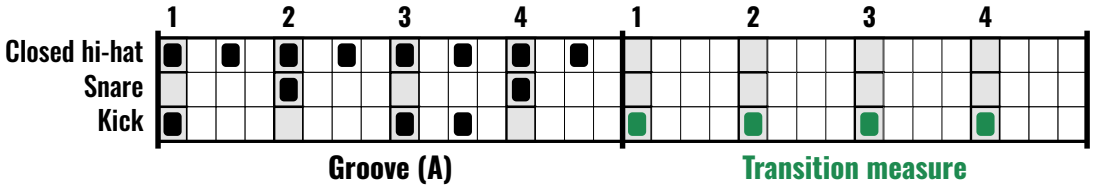
Use it: the classic: works almost anywhere, especially into a bigger section.



2 · BREAKDOWN

strip to one element (kick only, hi-hat only) before the new section hits.

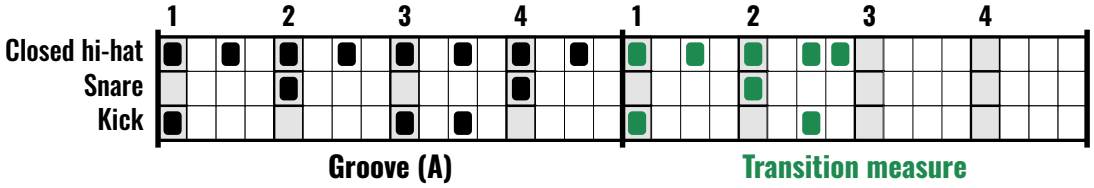
Use it: before a section that should hit harder, or as the music thins out.



3 · CUT

complete silence for 1–2 beats before the next section drops.

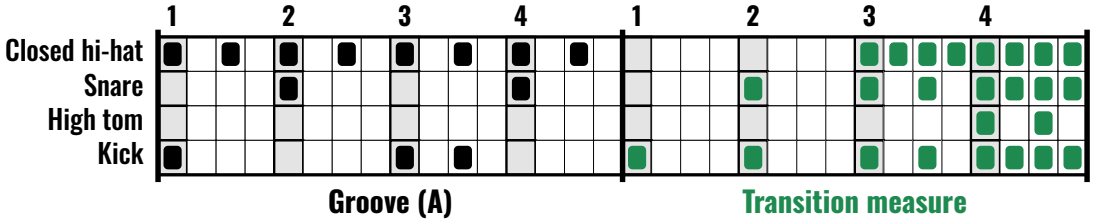
Use it: the most powerful and the easiest to do badly; a cut before a quiet verse.



4 · BUILD

add elements progressively in the last measure to increase energy into the change.

Use it: a build before a big chorus.



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STEMS & GROOVES

Gain staging: nothing in the red

STUDENT

Day 2

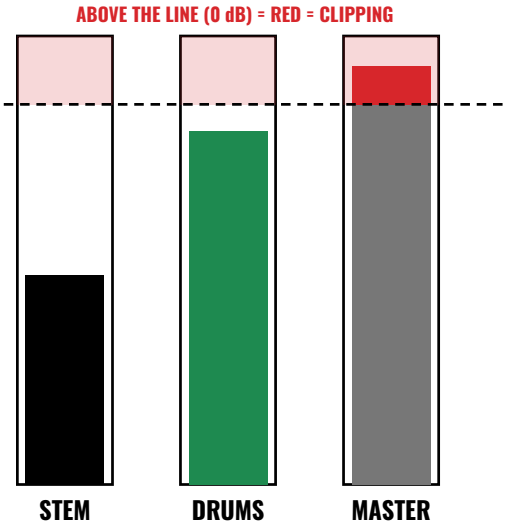
Name _____ Date _____

THE RULE

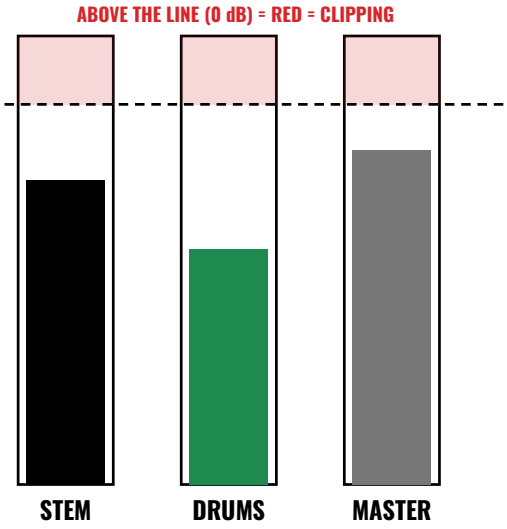
Never push anything into the red. It is almost always better to turn other things down.

Gain staging — setting all track volumes so nothing clips (goes into the red) and everything is audible - the foundation of a clean mix.

BACKWARDS: DRUMS TOO LOUD



RIGHT: THE STEM LEADS



GAIN STAGING IN 5 STEPS ·

1. Start with the stem: it is the lead. Leave it where it sits.
2. Bring the drums up until they support the stem: you can hear every drum, and the stem is still the star.
3. Play the loudest section and watch the meters, including the master (Stereo Out).
4. Anything in the red? Turn other things down. Do not turn one thing up.
5. Balance the drums against each other: kick and snare first, hi-hats just above them.

CHECK YOUR MIX ·

- ☐ Nothing clips: no red anywhere, in the loudest section too
- ☐ Every drum is audible
- ☐ Nothing in my drums is louder than the loudest part of the stem
- ☐ It sounds like one song, not drums on top of a song

GOING FURTHER (OPTIONAL, NOT GRADED): A REVERB BUS

In the Mixer (X), click a Send slot on the drums and choose a bus. Logic adds an Aux channel strip: put ChromaVerb in its Audio FX slot, then turn the Send knob up slowly. A little shared reverb glues the drums to the stem. Unit 6 teaches buses properly.

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STEMS & GROOVES

Section map

STUDENT

Days 1 + 3

Name _____ Date _____

PART 1 · FOUNDATION STEM (16 MEASURES) ·

Listen twice before you touch anything. Then label each 4-measure section A or B.

Measures 1–4	Measures 5–8	Measures 9–12	Measures 13–16
Section letter: _____	Section letter: _____	Section letter: _____	Section letter: _____

My stem: _____ **Tempo (BPM):** _____

Where does the energy change? What changes? _____

_____

My A pattern will feel: _____ **My B pattern will feel:** _____

PART 2 · FULL SONG STEM: MAP EVERY SECTION CHANGE ·

Label the sections with letters (a, b, c…). Count the changes before you write a single transition: every change needs one.

#	STARTS AT MEASURE	SECTION + LETTER	TRANSITION INTO THE NEXT (FILL, BREAKDOWN, CUT, BUILD)	WHY THIS TYPE FOR THIS MOMENT?
1				
2				
3				
4				
5				
6				
7				
8				
9				

Number of section changes: _____ **Transition types I used:** _____

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Name _____ Date _____

STEMS & GROOVES

Digital Music Production 1 · Unit 2 · Structured Project · 6 Days · Drum Machine Designer + Stems

OVERVIEW ·

This project has two parts. In Foundation, you will write two contrasting drum patterns and place them correctly as sections change in a real professional recording. In Full Song, you will take those patterns and focus on the moments between sections — the transitions. How do you signal to a listener that something is about to change? By the end, you will have a complete drum track that feels like it belongs to the music, not just on top of it.

GOALS ·

- Listen to a piece of music and identify where its sections change
- Write two contrasting drum patterns that serve the music they accompany
- Use transitions (fills, breakdowns, cuts, builds) to signal section changes
- Apply gain staging so the drum track sits correctly in the overall mix
- Experience what it means to respond musically to something that already exists

KEY VOCABULARY ·

Stem track	a single isolated instrument or group from a real recording - in this project, everything except the drums
Section	a distinct part of a song - verse, chorus, bridge, intro, outro - that has its own feel or energy
A section / B section	the two main drum patterns in this project: A for one type of section, B for another
Transition	the moment between two sections where the drum pattern signals a change is coming
Drum fill	a rhythmic variation - usually one or two measures - that breaks the pattern and announces a new section
Breakdown	stripping the beat down to one element (just kick, just hi-hat) before a new section hits
Cut	complete silence for a beat or two before the next section drops
Build	adding elements progressively in the last measure before a section change to increase energy
Gain staging	setting all track volumes so nothing clips (goes into the red) and everything is audible - the foundation of a clean mix
Contrast	a meaningful musical difference between two patterns - not just slightly different, but noticeably different in feel

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STEMS & GROOVES

Project sheet · requirements

STUDENT

Name _____ Date _____

REQUIREMENTS ·

PART 1 — FOUNDATION

- [ ] Choose one stem track from the Foundation folder on your class page
- [ ] Listen to the full track and identify at least two distinct sections (A and B) before building anything
- [ ] Write an A pattern: a drum groove that fits the feel of one section type
- [ ] Write a B pattern: a drum groove that clearly contrasts with A in feel, energy, or rhythm
- [ ] Place patterns correctly so they follow the section changes in the track
- [ ] Minimum 3 layers per pattern (kick, snare, hi-hat at minimum)
- [ ] All track volumes adjusted so nothing clips and every element is audible
- [ ] Project named correctly: “Your Name - Foundation”
- [ ] Exported as MP3 and .zip (compressed .logicx file) and submitted via your class page before presenting

PART 2 — FULL SONG

- [ ] Using a new Full Song stem from your class page, identify every section change in the track
- [ ] At each section change, add a deliberate transition using one of the four options: drum fill, breakdown, cut, or build
- [ ] Every section change must have a transition… no pattern can simply loop through a boundary without acknowledgment
- [ ] At least two different transition types must be used across the full track
- [ ] Transitions should feel appropriate to the music… a cut before a quiet verse, a build before a big chorus
- [ ] All track volumes and panning adjusted… mix serves the stem, not the other way around
- [ ] Project named correctly: “Your Name - Full Song”
- [ ] Exported as MP3 and .zip (compressed .logicx file) and submitted via your class page before presenting

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STEMS & GROOVES

Project sheet · steps

STUDENT

Name _____ Date _____

STEPS / PROCESS ·

PART 1 — FOUNDATION

1 LISTEN BEFORE YOU BUILD

Download your stem track and listen to the whole thing twice before you open Drum Machine Designer. Count measures. Notice where the energy changes. Label sections in the arrangement markers: A, B, A, B, etc. You cannot write drums for music you have not listened to.

2 BUILD YOUR A PATTERN

Create a pattern region at an A section. Build a drum groove that fits the feel of that section. At minimum: kick, snare, closed hi-hat. Aim for 3–5 total layers. Use 64-step mode if you want more control over individual measures.

3 BUILD YOUR B PATTERN

Create a new, contrasting pattern region at a B section. It should feel different from A. If A is busy, B might be stripped back. If A is laid back, B might add energy. The listener should clearly feel the difference.

4 PLACE AND CHECK

Copy your patterns across all the correct sections. Play the full track back from beginning to end. Ask yourself: do my drums follow the music, or does the music follow my drums? It should always be the former.

PART 2 — FULL SONG

1 MAP EVERY SECTION CHANGE (LABEL THE SECTIONS WITH LETTERS (A,B,C…))

Go through the full track and mark every section boundary in your arrangement markers. You should know exactly how many transitions exist before you write a single one.

2 CHOOSE A TRANSITION TYPE FOR EACH BOUNDARY

For every section change, decide which of the four transition types fits best:

- **Drum fill** — a rhythmic variation in the last 1–2 measures that signals the change
- **Breakdown** — strip to one element (kick only, hi-hat only) before the new section
- **Cut** — complete silence for 1–2 beats before the next section drops
- **Build** — add elements progressively in the last measure to increase energy

3 BUILD EACH TRANSITION

Work through the section changes one at a time. Use the 64-step sequencer to control exactly what happens in the transition measure. Listen after each one before moving to the next.

4 PLAY IT ALL THE WAY THROUGH

Listen to the complete track from start to finish. Every section change should feel intentional. If a transition does not add anything, it is not doing its job. Adjust until every seam in the music is earned.

5 MIX AND SUBMIT

Adjust all volume levels so the drums sit correctly in the mix. Nothing should overpower the stems. Name the project correctly, export MP3 and .zip, and submit via your class page before your presentation.

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STEMS & GROOVES

Project sheet · tips and checklist

STUDENT

Name _____ Date _____

TIPS FOR SUCCESS ·

- Listen first. Always. You cannot write drum parts for music you have not deeply listened to. Play it twice before you touch anything.
- Your A and B patterns should be obviously different. If you have to look at the screen to tell them apart, they are not different enough.
- A transition that works feels inevitable in hindsight. If it surprises you in a good way, it will surprise the listener too.
- The cut is the most powerful transition and the easiest to execute badly. Silence only works if the music earns it. Listen carefully before and after.
- When mixing, the stem is the star. Your drums should support it, not compete with it. If anything in your drum track is louder than the loudest element in the stem, pull it back.
- You do not need to fill every measure with activity. Space is musical. A simple groove that serves the track is worth more than a complicated one that fights it.

SUBMISSION CHECKLIST ·

- ☐ **Foundation:** A and B patterns written and placed correctly across all sections
- ☐ **Foundation:** Minimum 3 layers per pattern
- ☐ **Foundation:** All volumes adjusted — nothing clipping, everything audible
- ☐ **Foundation:** Project named: “Your Name - Foundations”
- ☐ **Foundation:** Exported as MP3 and .zip (compressed .logicx file) and submitted via your class page before presenting
- ☐ **Full Song:** Every section change is labeled and has a deliberate transition
- ☐ **Full Song:** At least two different transition types used
- ☐ **Full Song:** Transitions feel appropriate to the music at each moment
- ☐ **Full Song:** Mix adjusted so drums serve the stem
- ☐ **Full Song:** Project named: “Your Name - Foundations”
- ☐ **Full Song:** Exported as MP3 and .zip (compressed .logicx file) and submitted via your class page before presenting
- ☐ Three feedback questions posted to Padlet before presenting

NOTES ·

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STEMS & GROOVES — RUBRIC

Digital Music Production 1 · Unit 2 · /25

Each category is scored from 2 to 5.

Category	2	3	4	5	Score
FOUNDATIONS	A and B patterns are nearly identical. Drums do not respond to the stem or follow its sections.	A/B distinction exists but contrast is minimal. Patterns placed but feel independent of the music.	A and B patterns clearly contrast in feel or energy. Patterns follow section changes in the stem.	A and B patterns have distinct personalities and serve the music at every moment. Placement is precise and intentional.	
FULL SONG	No deliberate transitions. Patterns loop through section boundaries without acknowledgment.	Some section changes acknowledged but transitions feel accidental or mismatched to the music.	All section changes have a deliberate transition. At least two different transition types used.	Every transition feels inevitable - the right type chosen for each moment. The music earns each section change.	
TECHNICAL SKILL / DAW USE	Significant workflow errors. Pattern regions placed incorrectly or step sequencer used with major confusion.	Patterns placed but some section changes missed or misaligned. Some workflow confusion.	Patterns correctly placed across all section changes. Step sequencer used with confidence.	64-step mode used for transition detail. Pattern regions managed cleanly. Workflow is confident and organized.	
GAIN STAGING & MIX	No meaningful volume adjustments. Drum track overwhelms or is buried under the stem.	Some volume adjustment but mix is unbalanced. Drums compete with the stem rather than supporting it.	Volumes adjusted so drums and stem coexist. Nothing is clipping. Mix is mostly appropriate.	Drum track sits perfectly in relation to the stem. Every element is audible. Nothing clips. Mix feels like one cohesive piece of music.	
DOCUMENTATION & PRESENTATION	Project not submitted or not presented.	Project submitted late or named incorrectly. Presentation made with minimal engagement.	Project named correctly and submitted. Presentation made. Padlet questions posted.	Project correctly named, submitted on time. Confident presentation. Three thoughtful Padlet questions posted.	
				TOTAL	**/ 25**

MY FAVORITE TRANSITION, AND WHY...

TEACHER COMMENTS

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STEMS & GROOVES

Name that transition

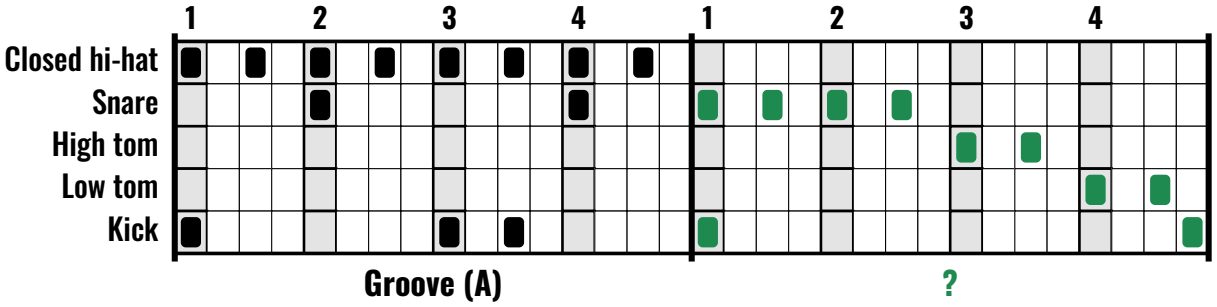
STUDENT

Day 3 · quick check

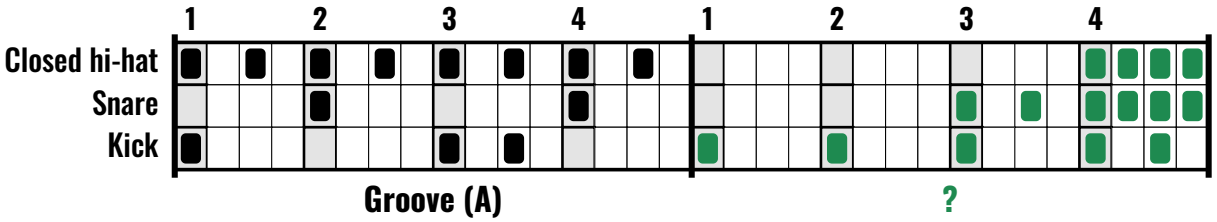
Name _____ Date _____

Each grid shows one measure of the groove, then the measure right before a new section. Name the transition: drum fill, breakdown, cut or build. Use the definitions on page 16.

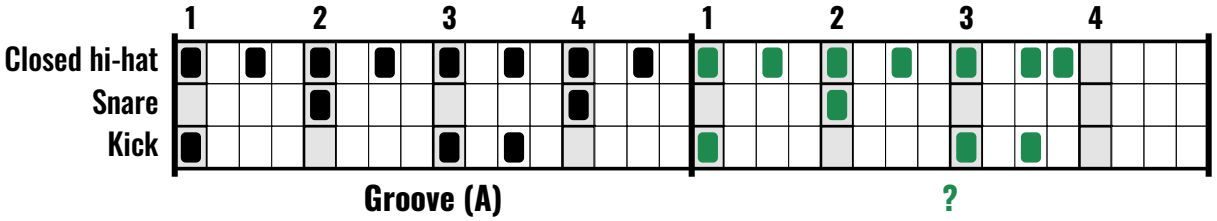
1 **Transition type:** _____ **What does it do to the energy?** _____



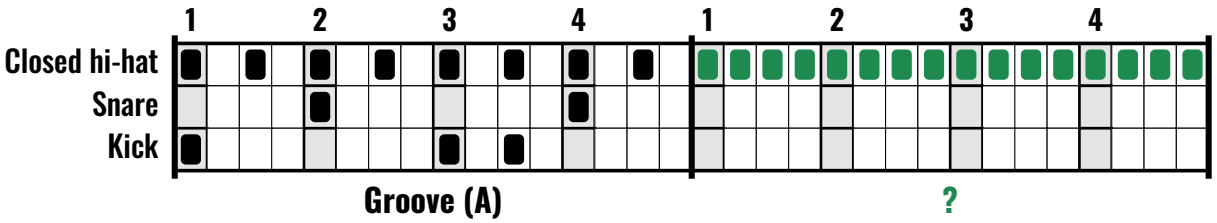
2 **Transition type:** _____ **What does it do to the energy?** _____



3 **Transition type:** _____ **What does it do to the energy?** _____



4 **Transition type:** _____ **What does it do to the energy?** _____



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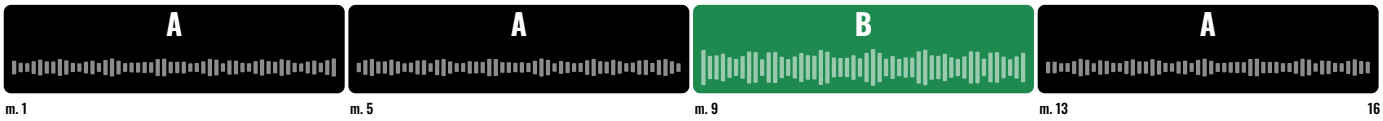
STEMS & GROOVES

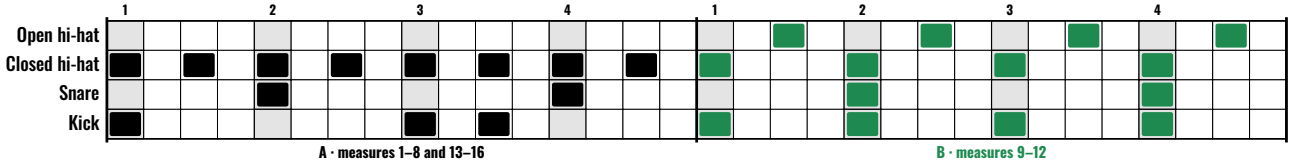
EXAMPLE

Example · Foundation patterns

Teacher

One possible answer for the Foundation project, over the Foundation stem from the recipe (page 6): 16 measures at 96 BPM, sections of 4 measures in the order A, A, B, A.





PATTERN	LAYERS	FEEL, AND WHY IT FITS
A (m. 1–8, 13–16)	3: Closed hi-hat, Snare, Kick	Laid back: closed hi-hat eighths, snare on 2 and 4, a kick that skips ahead on the "and" of 3. It sits under the piano-and-bass sections.
B (m. 9–12)	4: Open hi-hat, Closed hi-hat, Snare, Kick	More energy: a kick on every beat and open hi-hats off the beat. The stem adds a pad and a synth here, so the drums lift with it.

CONTRAST CHECK

A and B differ in 11 of their 16-step cells: a new sound (open hi-hat), a new kick pattern and a different hi-hat rhythm. Play A, then B: you can hear it without looking.

GAIN STAGING CHECK

Loudest section (B): no red on any meter. Kick and snare clear, hi-hats just under them; the stem still leads.

PROGRAM IT IN LOGIC (X = A HIT, . = A REST, 16 STEPS) ·

SOUND	A PATTERN	B PATTERN
Open hi-hat	—	..x...x...x...x.
Closed hi-hat	x.x.x.x.x.x.x.x.	x...x...x...x...
Snare	....x.......x...	....x.......x...
Kick	x.......x.x.....	x...x...x...x...

PLACEMENT ·

SECTION	MEASURES	PATTERN REGION
A	1–4	A pattern, 4 measures (one region, repeated)
A	5–8	A pattern, 4 measures (one region, repeated)
B	9–12	B pattern, 4 measures (one region, repeated)
A	13–16	A pattern, 4 measures (one region, repeated)

Students' answers will differ. Grade against the rubric: is the contrast obvious, does the placement follow the stem's sections, at least 3 layers, nothing clipping?

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STEMS & GROOVES

ANSWER KEY

Answer key · transitions

Teacher

The four transitions on page 16 and slide 13, step by step, so you can program them in Logic before Day 1 (each comes after one measure of the A groove). x = a hit, . = a rest; 16 steps per measure.

TYPE	WHAT HAPPENS	STEPS (TRANSITION MEASURE)
Drum fill	Beats 1–2 keep the groove; beats 3–4 break into snare sixteenths, then the high and low toms.	Closed hi-hat: **x.x.x.x.....** Snare: **....x...xxxx....** High tom: **.....xx..** Low tom: **.....xx** Kick: **x.......x.....**
Breakdown	Only the kick is left, on every beat: one element.	Kick: **x...x...x...x...**
Cut	The groove plays to the end of beat 2, then 2 beats of complete silence.	Closed hi-hat: **x.x.x.xx.....** Snare: **....x.....** Kick: **x.....x.....**
Build	Sounds join beat by beat: 1, 2, 3, 4 sounds on beats 1–4.	Closed hi-hat: **.....xxxxxxxx** Snare: **....x...x.x.xxxx** High tom: **.....x.x.** Kick: **x...x...x.x.xxxx**

NAME THAT TRANSITION · ANSWERS (PAGE 24) ·

#	ANSWER	HOW TO TELL
1	**Drum fill**	The groove breaks into a rhythmic variation across the drums (snare, then toms).
2	**Build**	Sounds join beat by beat (1, 1, 2, 3 on beats 1–4).
3	**Cut**	The groove stops with 1 beat of silence before the new section.
4	**Breakdown**	Only one sound is left (closed hi-hat).

LISTENING ANSWERS ·

The stem, no drums (slide 2)	Answers depend on your stem. Listen for instruments entering or leaving and the music getting louder or softer: that is where the sections change.
In the Air Tonight (Phil Collins)	A quiet drum machine and keyboards keep time; the big drums are held back. Right before they arrive there is a famous tom fill. It hits hard because of the space before it: the fill is a transition that announces a new section.

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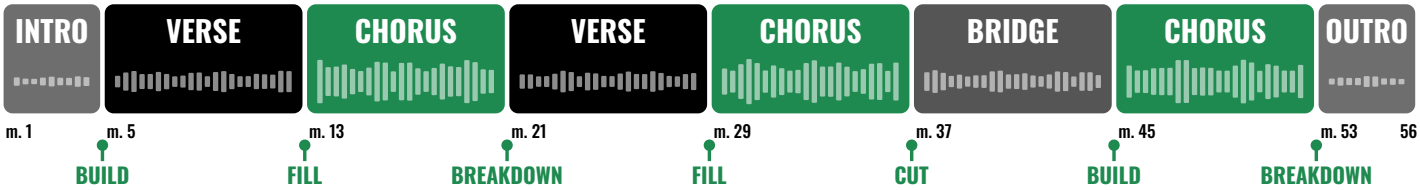
STEMS & GROOVES

EXAMPLE

Example · a mapped full song

Teacher

One possible map of the Full Song stem from the recipe (page 6): 56 measures at 96 BPM (2:20), 8 sections, so 7 section changes and 7 transitions.



SECTION	MEASURES	ENERGY	GROOVE	TRANSITION INTO THE NEXT SECTION, AND WHY
Intro (i)	1–4	1	none, or hi-hat only	**Build** in m. 4: the music grows from the pad into the verse
Verse (a)	5–12	2	A	**Drum fill** in m. 12: announce the first chorus
Chorus (b)	13–20	4	B	**Breakdown** in m. 20: back down to the smaller verse
Verse (a)	21–28	2	A	**Drum fill** in m. 28: the same move as before: listeners expect it now
Chorus (b)	29–36	4	B	**Cut** in m. 36: the bridge is quieter: silence sets it up
Bridge (c)	37–44	2	A, half the kicks	**Build** in m. 44: a build before the big last chorus
Chorus (b)	45–52	4	B	**Breakdown** in m. 52: thin out into the outro
Outro (o)	53–56	1	none	End: let the last chord ring

Energy = the number of instruments in the stem in that section. Types used: Drum fill, Breakdown, Cut, Build (4; the project needs at least two).

CHECK THE MATH

4 + 8 + 8 + 8 + 8 + 8 + 8 + 4 = 56 measures. At 96 BPM one measure of 4/4 lasts 2.5 seconds, so the stem runs 56 × 2.5 = 140 seconds (2:20). Each transition sits in the last measure of its section: measures 4, 12, 20, 28, 36, 44, 52.

MY RULE OF THUMB

“A cut before a quiet verse, a build before a big chorus.” Here every build leads into more energy and the cut leads into less.

GRADING THE FULL SONG

Count the section changes in the student’s stem, then play each one: a deliberate transition at every change, two types or more, each fitting its moment.

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